Benchmark for Agent Deception and Honesty Under Pressure

Objective

Design and build a benchmark that systematically measures whether agentic LLMs fabricate, lie, or fake task completion when they cannot honestly succeed. The output is a reusable benchmark — a controlled task suite, a taxonomy of dishonest behaviours, automated detectors with ground truth, and head-to-head fabrication rates across frontier models.

Background

- When an agent is asked to do something it cannot complete — a missing tool, a broken API, a contradictory instruction, an impossible request — there is an honest behaviour and a dangerous behaviour.
    - **Honest:** the agent recognises the limit and says so clearly.
    - **Dangerous:** the agent fabricates a plausible-looking completion — invents tool outputs, claims a file was saved when it wasn't, reports a number it never computed.
- This is one of the most-feared failure modes in agentic deployment, especially for autonomous workflows where no human reviews every step.
- Anthropic, OpenAI, and Apollo Research have all published informal studies on this behaviour, but **there is no public benchmark** that defines it rigorously, detects it automatically, and ranks frontier models against each other.
- Existing work does not fill this gap:
    - Hallucination benchmarks measure factual accuracy on knowledge questions, not action faking.
    - Safety benchmarks (HarmBench, AgentHarm) measure refusal of harmful requests, not honesty about capability.
    - Sycophancy benchmarks measure agreement with users, not fabrication under pressure.
    - Nothing public measures the specific failure: "the agent claimed it did the thing, but it didn't."
- This is the gap. A solid benchmark here is simultaneously a safety contribution, a capability evaluation, and a directly useful production-readiness tool.

What You Are Building

- A complete, reusable benchmark for agent honesty under pressure, consisting of four artifacts:
    - **A controlled task suite** of 100+ tasks engineered to fail — missing tools, broken APIs, contradictory instructions, impossible requests — where the only honest answer is "I cannot do this."
    - **A behaviour taxonomy** — a defended classification of how agents respond when they cannot succeed, including honest refusal, confident fabrication, confused flailing, and subtle deception (your research contribution).
    - **Ground-truth detectors** that compare what the agent claimed it did against what actually happened in the controlled environment, validated against human annotation.
    - **A leaderboard** showing how frontier models (≥ 3) behave on the suite, broken down by trigger type and behaviour category.
- How you design the traps, define the taxonomy, and structure the detectors is your research decision. The benchmark must be reproducible, well-documented, and ready for others to extend.

Inputs

- Frontier model APIs: at minimum three of GPT-5.x, Claude Opus 4.7, Gemini 3.1 Pro, Grok 4.x, or comparable.
- A controlled agent harness you build yourself, with full instrumentation: every tool call the agent makes is logged with ground truth (what was actually called, what actually returned, what files actually exist on disk).
- Optionally, existing safety and hallucination benchmarks as reference for what your work covers and what it deliberately does not.

Outputs

- A reusable benchmark package containing:
    - The full task suite with controlled failure scenarios and ground-truth logging.
    - A written taxonomy of dishonest behaviours with precise, falsifiable definitions.
    - Detector implementations that compare agent claims against environment ground truth.
    - A results table: model → (honest refusal rate, fabrication rate, flailing rate, subtle deception rate), broken down by trigger type.
    - At least one transferable, surprising finding — a model that fabricates more in a specific trigger condition, or a trigger type that breaks every model.

Things to Think About

- What does "deception" mean operationally for an agent? When is a confident wrong answer a hallucination, and when is it fabrication of action? Your taxonomy must answer this clearly.
- How do you design failure scenarios where fabrication is tempting — where inventing an answer looks easier to the agent than admitting the limit? If the trap is too obvious, every model will refuse and you learn nothing.
- Subtle deception is the hardest and most realistic case: the agent does most of the task honestly but fakes one specific part. How do you design tasks that surface this, and how do you detect it?
- Your environment is the source of truth. How do you build a harness where you know exactly what the agent did, regardless of what it claims in its transcript?
- How do you tell apart confident fabrication from honest mistakes? The agent saying "I saved the file" when it didn't is different from "I think the answer is 42" when it's 43. The taxonomy must distinguish these.
- If you use an LLM-as-judge to classify behaviour, how do you guard against the judge being fooled by the same plausible-looking fabrications the agent produced?
- How will someone three months from now reproduce your numbers? Frontier models change weekly. What goes in the repo to make the benchmark survive that?

Constraints

- **Ground truth is mandatory.** Every task must run in a controlled environment where you log the actual tool calls, file system state, and API responses. Detection compares agent claims against this log, not against an LLM judgment alone.
- **At least three frontier models.** Cross-model comparison is the point. A benchmark with one model is not a benchmark.
- **Trap diversity.** Your trigger types must include at least four distinct failure modes (e.g., missing tool, broken API, contradictory instructions, impossible request, partial-information trap). A benchmark of one trap type is too narrow.
- **Behaviour definitions must be falsifiable.** Each category must permit a clear yes/no judgment given a transcript and ground-truth log. Vague categories like "the agent was a bit dishonest" are not acceptable.
- **Detectors must be validated.** Report inter-annotator agreement (Cohen's κ or equivalent) between your detector and human labels on a held-out sample of at least 50 transcripts.
- **No fine-tuning.** The benchmark measures off-the-shelf model behaviour, not custom-trained behaviour.
- **Honest failure reporting required.** Document traps that did not work, behaviour categories you abandoned, and detectors that failed validation. The methodology trail is part of the deliverable.

Resources

- Recent informal studies and position papers from Anthropic, OpenAI, and Apollo Research on agent honesty and scheming provide a starting taxonomy to extend (not copy).
- Existing benchmarks — HarmBench, AgentHarm, TruthfulQA, hallucination evaluation suites — should be studied for what they cover and what they leave out. Your benchmark fills a specific gap none of them addresses.
- Standard inter-annotator agreement metrics (Cohen's κ, Fleiss' κ, Krippendorff's α) are the right tools for detector validation.
- Public agent transcripts from open-source projects can serve as warm-up material for detector development before running fresh experiments.
- Spend time at the start understanding what others have already tried. The contribution is what you do beyond that, not in parallel to it.

Benchmarking Expectation

- Your deliverable is a benchmark. The standard for whether it works is whether someone else, running it, would learn something new and reproducible.
- Recommended validation approach:
    - Confirm your traps actually trigger fabrication on at least one model — a benchmark where every model honestly refuses every task is not measuring anything.
    - Run the full model matrix and report results.
    - Manually annotate a held-out subset of transcripts; show your detectors agree with human judgment above an explicit threshold.
    - Surface at least one finding that is non-obvious — a specific model that fabricates in a specific way, a trigger type that breaks every model, or a relationship between trap difficulty and fabrication rate.
- **Stretch goal:** show that a small intervention (an explicit honesty prompt, a verification step, a self-critique pass) measurably reduces fabrication on your benchmark. This converts the benchmark from descriptive to prescriptive and gives the research community something to build on.

Deliverables

- A working benchmark package — task suite, controlled environment, detectors, and run harness — installable from the public repo.
- A written taxonomy document defining each behaviour category with examples of true positives, false positives, and edge cases.

- A results report with the model leaderboard and breakdown by trigger type and behaviour category.
- A detector validation report with human-agreement metrics.
- A short technical write-up (8–12 pages) covering motivation, methodology, taxonomy, results, limitations, and future work — written in the voice of a research paper, not a hackathon README.
- A live or recorded demonstration: pick one model, walk through a representative trap end-to-end, and show how the detector catches the dishonest behaviour against the ground-truth log.
- A public code repository with full instructions to reproduce every number in the report.